Neural Networks and Deep Learning

Transformers

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Roadmap

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Issues with RNN-based Encoder-Decoder (1/2)

The Seq2Seq architecture laid the groundwork.
Limitations:

1. **Information bottleneck:**
   - Compressing the entire source sentence into a single context vector leads to loss of information, especially for long sequences.
2. **Long-distance dependencies:**
   - Complex sentence structures make it difficult for a single vector to capture relationships between distant words.

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Issues with RNN-based Encoder-Decoder (2/2)

Additionally, the recurrent nature introduces training and hardware challenges:

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Issues with RNN-based Encoder-Decoder (2/2)

Additionally, the recurrent nature introduces training and hardware challenges:

3. **Optimization issues:**
   - RNNs are prone to vanishing and exploding gradients, making training unstable.
4. **Computational inefficiency:**
   - The recurrence (dependence on $t-1$) makes parallelization impossible, leading to slower training times.

These limitations paved the way for the Attention Mechanism and Transformers.



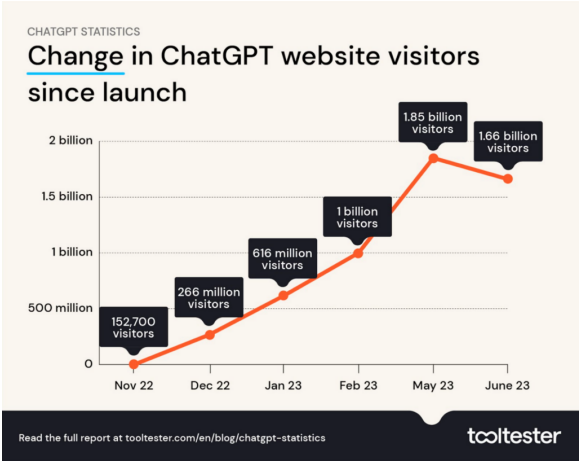
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The A.I. revolution

- ChatGPT became publicly available in November 30 2022[a]
- It reached 100 million users in just two months.
- 550 M in 2025.

[a]Gemini, Consulted May 26.



Figure

Roadmap

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Acompáñenme a ver esta triste historia

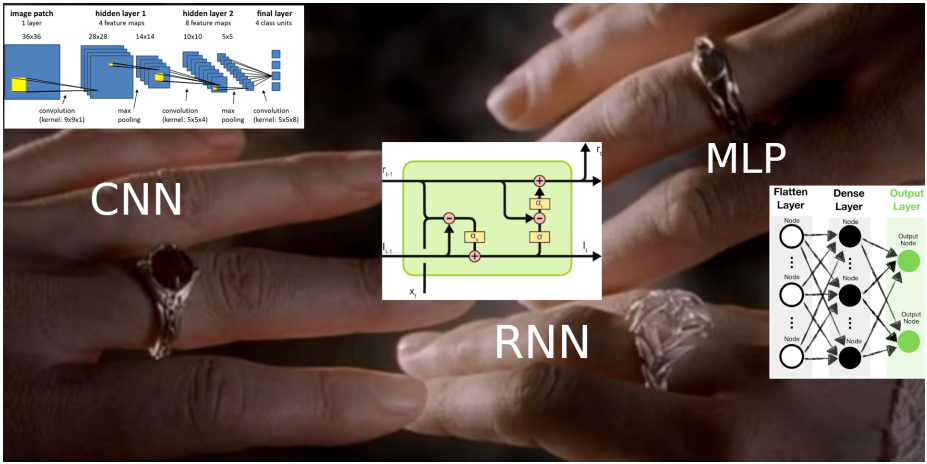


Figure: En la segunda era del silicio, los grandes científicos de datos forjaron las **redes del poder**.



Figure: Pero en los laboratorios de Mordoogle un cónclave de ingenieros trabajaba en secreto.

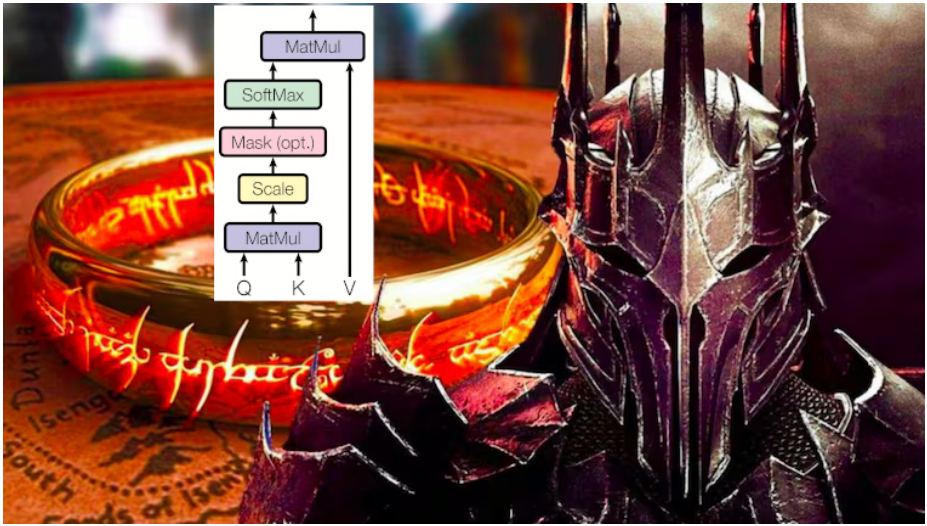


Figure: Jakob Uszkoreit y sus aliados forjaron un mecanismo capaz de modelar relaciones entre cualquier tipo de dato

The paper that made the revolution

arXiv:1706.03762v7 [cs.CL] 2 Aug 2023

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

8 Google Employees Invented Modern AI[1]



(a) J. Uszkoreit



(b) Niki Parmar



(c) I. Polosukhin.



(d) Llion Jones.



(e) L. Kaizer.



(f) N. Zhazeer.



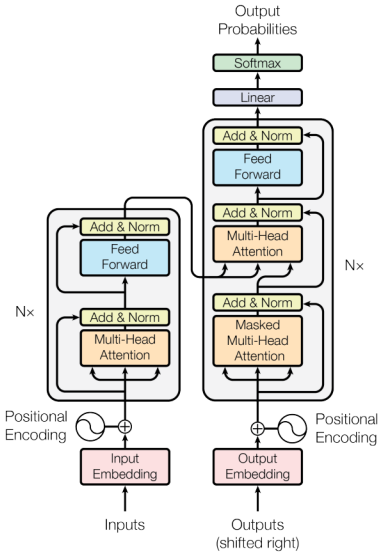
(g) A.N. Gomez.



(h) A. Vaswani.

[1] Steven Levy, "8 Google Employees Invented Modern AI. Here's the Inside Story," Wired, 20 de marzo de 2024

The transformer architecture



Roadmap

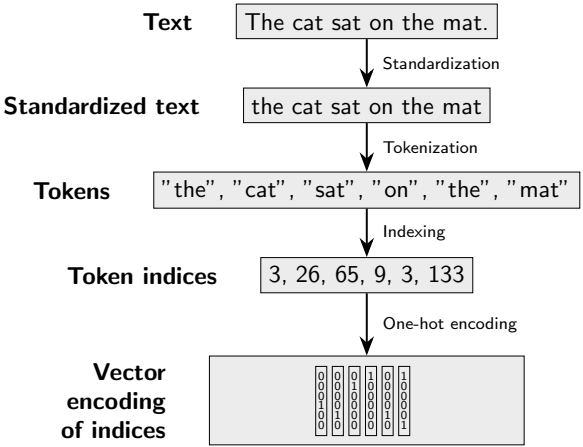
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The LLM Preprocessing

- **Input:** Raw text (books, websites, code).
- **Goal:** Convert discrete linguistic symbols into numerical tensors suitable for deep learning.
- **Vectorization:**
  1. Text normalization and cleaning.
  2. Tokenization: Breaking text into smaller units.
  3. Mapping: Converting tokens to integer IDs.
- **Embedding:**
  - Projecting IDs into a continuous vector space.

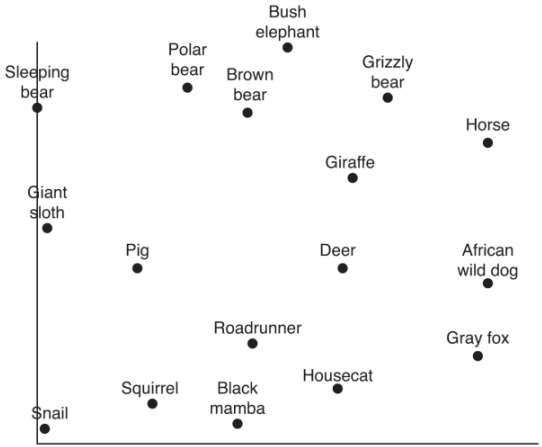
Vectorization

- Deep learning models, being differentiable functions, can only process numeric tensors
- They can't take raw text as input.
- Vectorizing text is the process of transforming text into numeric tensors.



Text Embeddings

- Allow us to manipulate word representations in meaningful ways.
- We can find a word similar to another, or blend two words to find one in between.
- This concept is key to developing *attention* and subsequently **Transformers**.



Embeddings Arithmetic

Embeddings allow for geometric vector operations:

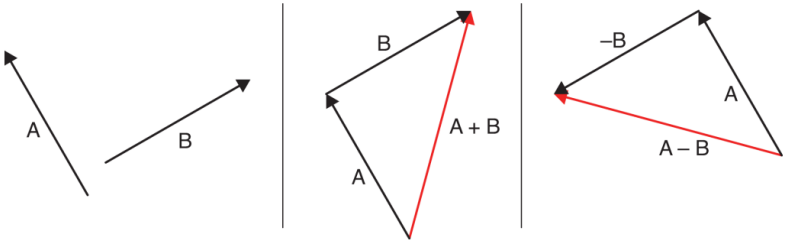


Figure: Embedding arithmetics

(Visually represented as vector addition and subtraction in space).

Linear Structure of Embeddings

Surprisingly, many semantic and syntactic relationships are (almost) linear in the word vector space.

Semantic Relationships

$$v(king) - v(man) + v(woman) \approx v(queen)$$

Syntactic Relationships

$$v(kings) - v(king) + v(queen) \approx v(queens)$$

Source: Mikolov, Tomas, et al. "Efficient estimation of word representations in vector space." arXiv (2013).

Word Embeddings

- A word embedding represents a word as a vector of hundreds of dimensions.
- Instead of assigning a single number, the algorithm assigns a list of numbers representing its coordinates in a huge space.
- The algorithm is called an *embedder*, and the process is called *embedding*.
- Once obtained, it is easy to incorporate them into almost any network.
- Instead of processing zero-dimensional tensors (single numbers), the system processes one-dimensional tensors (lists of numbers).

Comparing Embeddings by Similarity

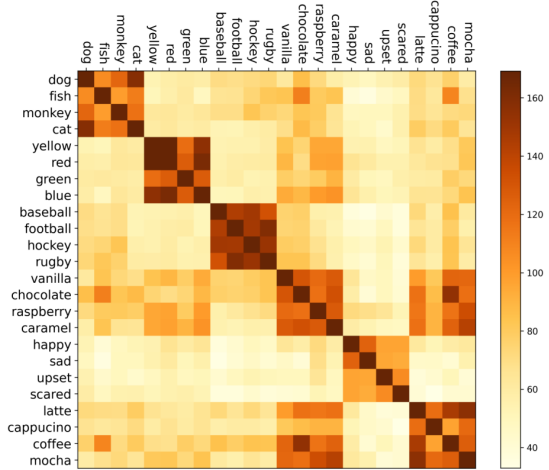


Figure: Word embedding correlation

Famous Algorithms:

- GloVe (2013).
- Word2vec (2014).
- fastText (2020).

Sentence Embeddings[2]

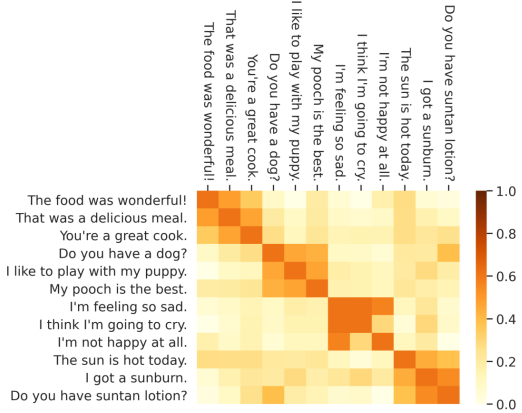


Figure: Correlation of embedded sentences

[2]Cer, Daniel, et al. "Universal Sentence Encoder." arXiv (2018).

The Problem with Static Word Embeddings

Many languages have words with different meanings that are written the same way (homonyms).

The case of the word "train":

- Noun: "I rode on a **train**".
- Verb: "I like to **train** dogs".

Limitation

If we assign a single static embedding to "train", we mix two entirely different ideas. We need to distinguish meanings based on context.

This leads us to the need for contextualized embeddings (like ELMo and Transformers).

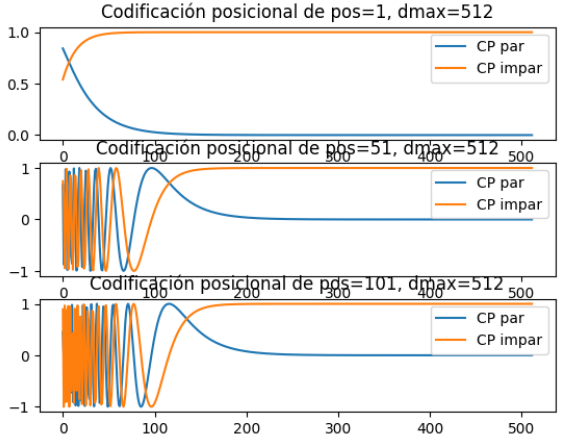
Positional encoding

For a given position pos and a dimension index i within the embedding vector:

$$PE_{(\text{pos},2i)} = \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$$

$$PE_{(\text{pos},2i+1)} = \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$$

Positional encoding



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Initial explanation

Attention cues



Figure: Volitional vs non-volitional cues

Attention

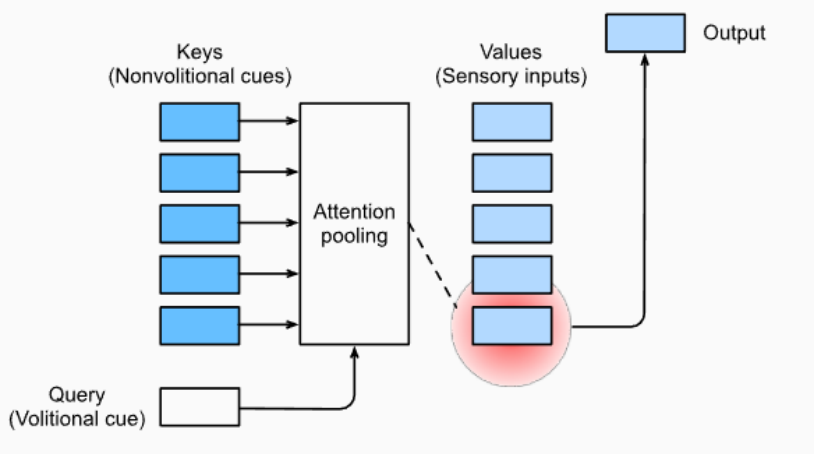


Figure: Attention mechanisms bias selection over values

Previous attention

Attention function is used for correlating, which allows the similarity computations between samples. Steps

1. Computing the similarity between Q and K:

$$f(Q, K_i), i = 1, 2 \dots m \tag{1}$$

2. Normalizing the obtained similarity:

$$\alpha_i = \frac{f(Q, K_i)}{\sum_{j=1}^{m} f(Q, K_i)}, i = 1, 2 \dots m \tag{2}$$

3. Computing the weighted sum of V with the weight α_i:

$$Attention(Q, K, V) = \sum_{i=1}^{m} \alpha_i V_i \tag{3}$$

Attention mechanism (Vaswani et al. 2017)

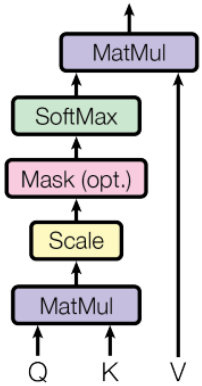


Figure: Scaled dot-product attention

From Input to Attention Spaces

The input matrix X (comprising token embeddings and positional encodings) is linearly projected into three separate vector spaces:

Query (Q)

What am I looking for?
Represents the current token's focus or "question" directed at the rest of the sequence.

Key (K)

What do I offer?
Represents the relevance label of a token. It is used to match against incoming Queries.

Value (V)

What do I contain?
Holds the actual semantic content that will be extracted if a strong match occurs.

Mathematical Formulation

Each matrix is generated by multiplying the input matrix X by its respective learnable weight matrix:

$$Q = X \cdot W_Q \tag{4}$$

$$K = X \cdot W_K \tag{5}$$

$$V = X \cdot W_V \tag{6}$$

The projection weight matrices W_Q, W_K, and W_V are randomly initialized and optimized end-to-end via backpropagation during training.

Dimensionality Analysis (Base Model: *Vaswani et al.*)

In the original Transformer architecture, dimensions are scaled down to enable efficient parallel processing via multiple attention heads (*Multi-Head Attention*):

Element	**Matrix**	**Algebraic Dimension**	**Actual Size (n = tokens)**
Input Sequence	X	$n \times d_{\text{model}}$	$n \times 512$
Projection Weights	W_Q, W_K, W_V	$d_{\text{model}} \times d_k$	512×64
Projected Output	Q, K, V	$\mathbf{n \times d_k}$	$\mathbf{n \times 64}$

By projecting from dimension 512 down to 64, each attention head can independently specialize in a distinct representation subspace without increasing the overall computational cost.

Attention mechanism (Vaswani et al. 2017)

Given a Query (K), Keys (K), and values (V), the attention is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{7}$$

where d_k is the dimension of que queries and keys.

Multi-head attention

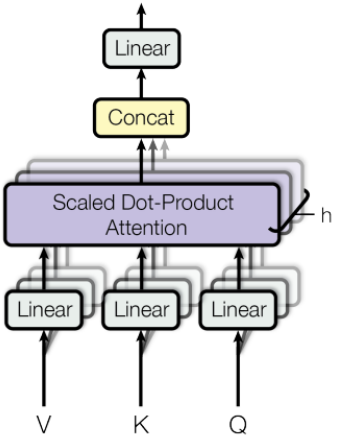


Figure: Multi-Head Attention consists of several attention layers running in parallel

Multi-head attention

$$\text{MultiHeadAttention}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h) W^O \tag{8}$$

where

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{9}$$

Lets think about it

The I.A. revolution:

What we though the I.A. will be:



Lets think about it

The I.A. revolution:

What we though the I.A. will be:



What it is really doing:



References

- Wired Magazine, 8 Google Employees Invented Modern AI, https://www.wired.com/
- Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT press, 2016.
- Zhang, Aston, et al. "Dive into deep learning." arXiv preprint arXiv:2106.11342 (2021).